LAB REPORT: ARP & ICMP USING PACKET TRACER

Title---

Study of ARP and ICMP using Simulation Mode in Cisco Packet Tracer.

Objective

- To observe packet flow in a LAN.
- To understand ARP and ICMP behavior.
- To analyze switch MAC address learning.

Tools Used

- Cisco Packet Tracer.

Theory

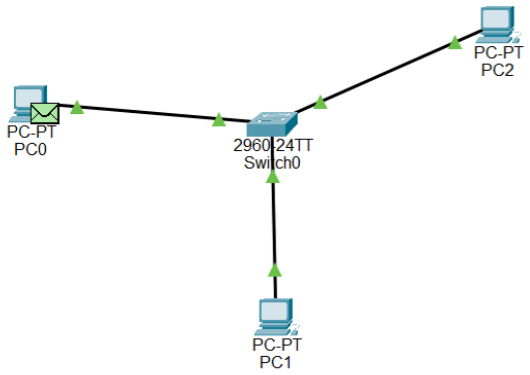
- **ARP (Address Resolution Protocol):**
  Converts IP address into MAC address using broadcast.
- **ICMP (Internet Control Message Protocol):**
  Used for communication testing (ping).
- **Switch MAC Learning:**
  Switch stores MAC addresses and forwards frames efficiently.

Procedure

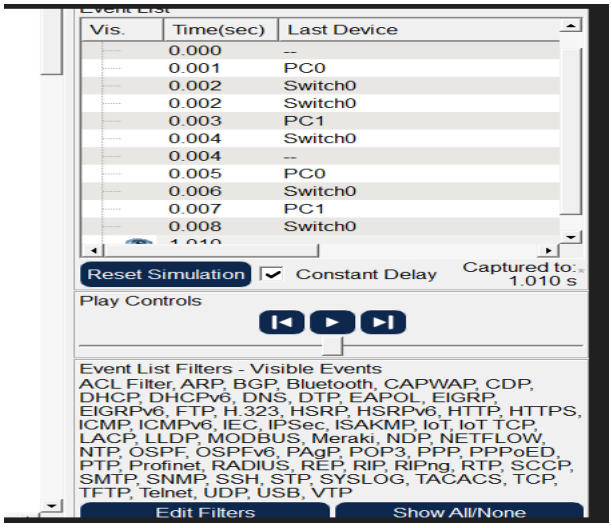
1. Created a LAN with 1 switch and 3 PCs .
2. Assigned IP addresses (192.168.20.x) .
3. Enabled Simulation Mode .
4. Performed ping from PC0 to PC1.
5. Observed ARP and ICMP packets .
6. Repeated ping to observe ARP caching.
7. Checked MAC address table .

Observations-----

This shows a LAN with 3 PCs connected to a switch.



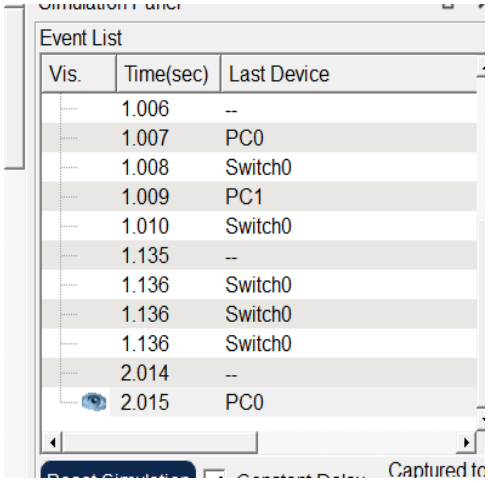
Simulation Mode Enabled



Simulation mode allows step-by-step visualization of packet flow.

ARP Request (Broadcast)

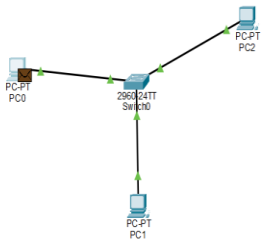
PC0 sends an ARP request to find the MAC address of PC1. The switch broadcasts this packet to all devices.

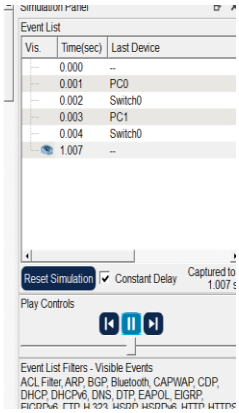


Event List

Vis.	Time(sec)	Last Device
	1.006	--
	1.007	PC0
	1.008	Switch0
	1.009	PC1
	1.010	Switch0
	1.135	--
	1.136	Switch0
	1.136	Switch0
	1.136	Switch0
	2.014	--
	2.015	PC0

Second Ping (No ARP)





Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	PC1
	0.004	Switch0
	1.007	--

Reset Simulation | Constant Delay | Captured to 1.007 s

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP,

. ICMP Echo Request—

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=8ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>
```

ICMP Echo Reply—

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=8ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>
```

MAC Address Table

The switch dynamically learns MAC addresses and maps them to ports for efficient communication.

```
   1    00d0.58b6.6e7b    DYNAMIC     Fa0/2
   1    00e0.a35c.66b7    DYNAMIC     Fa0/3
Switch#
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.961d.c947    DYNAMIC     Fa0/1
   1    00e0.a35c.66b7    DYNAMIC     Fa0/3
Switch#
```